

NABEEL ABDUL AZIZ KHAN

AI Engineer | Machine Learning Engineer | Generative AI Developer

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SUMMARY

AI Engineer with hands-on experience designing and deploying end-to-end intelligent systems, including LLM-powered applications, conversational agents, and real-time voice AI solutions. Experienced in building retrieval-augmented **pipelines**, agent-based workflows, and **context-aware systems** using **LangGraph** and **LangChain**. Strong foundation in backend engineering with **Python (FastAPI)** and **.NET**, enabling seamless integration of AI capabilities into scalable production systems. Focused on developing **robust AI workflows**, **prompt-driven systems**, and **data-driven automation solutions**, with emphasis on clean architecture and real-world deployment.

CORE TECHNICAL SKILLS

- **Generative AI & NLP:** LLMs, LangChain, LangGraph, RAG, Prompt Engineering, Conversational AI
- **Programming:** Python, C#, JavaScript, TypeScript, SQL
- **Frameworks & Libraries:** FastAPI, ASP.NET Core, Scikit-learn, Pandas, NumPy
- **AI Systems:** Speech-to-Text, Text-to-Speech, AI Agents, Workflow Automation
- **Cloud & Tools:** Azure, AWS (EC2, S3), Docker, Git, CI/CD
- **Databases:** PostgreSQL, SQL Server, MySQL, Pinecone
- **Vector Search & Retrieval:** Pinecone, embeddings, semantic search.

EDUCATION

Bachelor of Engineering (Computer Science Engineering) – Lords Institute of Engineering & Technology (2024)

Diploma in Computer Engineering – State Board of Technical Education

EXPERIENCE

Software Developer | [Vision Technologies](#) (Jan 2025 – Present)

- Designed and developed scalable applications using **ASP.NET Core and Angular**, contributing to modular and maintainable system architecture.
- Built **LLM-powered workflows using LangChain**, implementing context-aware response generation and document/data processing pipelines for automation use cases.
- Developed **retrieval-based systems using RAG architecture**, enabling dynamic querying of structured and unstructured data for intelligent responses.
- Built and maintained **REST APIs using .NET Web API**, integrating backend systems with AI pipelines for real-time processing.
- Optimized data handling in **SQL Server**, improving performance through indexing strategies and efficient query design.
- Contributed to designing **end-to-end AI workflows**, connecting backend services, LLM pipelines, and database systems for production-ready solutions.

Software Developer | [TekHQs Inc](#) (Oct 2021 – Dec 2024)

- Developed web applications using **Angular and ASP.NET MVC**, focusing on modular architecture and reusable components.
- Built backend services using **ASP.NET Web API and FastAPI**, supporting data processing and integration with AI-powered features.
- Implemented **LangChain-based pipelines** to process user inputs, retrieve relevant context, and generate structured LLM responses.
- Designed and implemented **RAG pipelines using Pinecone**, enabling semantic search over domain data and improving response relevance.

- Worked with **vector embeddings and similarity search**, enabling intelligent querying and contextual retrieval in AI systems.
- Developed and managed data systems using **SQL Server and PostgreSQL**, supporting both transactional operations and AI-driven workflows.
- Assisted in deploying applications on **Microsoft Azure**, supporting cloud-based hosting and service integration.

PROJECTS

AI Voice Receptionist for Car Dealership

- Designed and developed a real-time AI voice assistant using **Python**, handling orchestration of conversational workflows and backend logic.
- Built **stateful conversational workflows using LangGraph**, managing session state, multi-step interactions, and conversation flow control.
- Implemented **LangChain pipelines** to process user queries, retrieve relevant context, and generate dynamic responses using LLMs.
- Developed a **real-time voice pipeline using Whisper (Speech-to-Text), GPT-based LLMs, and Edge TTS (Text-to-Speech)** to enable seamless voice interactions.
- Integrated **LiveKit (WebRTC)** to support low-latency audio streaming and real-time communication between user and AI system.
- Connected **PostgreSQL** to dynamically fetch customer data and personalize responses based on conversation context.
- Designed an **end-to-end AI workflow (STT → LLM → TTS)** integrating voice processing, backend APIs, and data retrieval systems.

GitHub: [MediBot – AI Medical Chatbot](#)

- Developed an AI-powered chatbot using **Python, NLP, and OpenAI LLMs**, enabling symptom-based guidance and conversational interaction.
- Implemented **Retrieval-Augmented Generation (RAG)** using **LangChain and Pinecone**, enabling semantic search over medical knowledge bases.
- Built backend APIs using **FastAPI**, enabling real-time query handling and integration with frontend or external systems.
- Designed **prompt engineering workflows**, including structured prompts and validation layers to improve response accuracy and reduce hallucinations.
- Generated vector embeddings using **HuggingFace Embeddings (Hugging Face models)** and implemented similarity search with **Pinecone** to enable context-aware retrieval and response generation.
- Deployed the application on **AWS (EC2, S3)** and used **Docker** for containerized, scalable deployment.

GitHub: [AI Document Processing System](#)

- Built an intelligent document processing system using **Python**, enabling extraction of structured data from unstructured documents such as PDFs and images.
- Integrated **OCR tools (Tesseract / Azure Form Recognizer)** to extract raw text from documents for downstream processing.
- Implemented **LangChain-based LLM pipelines** to parse, analyze, and extract key information from documents.
- Designed **RAG workflows using vector embeddings and Pinecone**, enabling querying and contextual analysis of document data.
- Developed backend APIs using **FastAPI** to handle document upload, processing, and response delivery.
- Stored processed outputs in **PostgreSQL**, enabling structured storage, analytics, and reporting workflows.
- Designed a **scalable pipeline architecture** combining OCR, LLM processing, and database storage for handling large volumes of documents.